

MOTOR HOUSING COMPONENT " TECHNICAL SPECIFICATION

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1. SCOPE

This specification defines requirements for the precision motor housing assembly used in the Series 400 electric motor product line.

All suppliers must satisfy ALL requirements herein before shipments begin.

2. APPLICABLE STANDARDS

- ISO 2768-1 (General tolerances " linear and angular dimensions)
- ISO 1302 (Surface texture indication)
- ASTM B209 (Aluminum alloy sheet and plate)
- IPC-A-610 (Acceptability of electronic assemblies)
- RoHS 2011/65/EU (Restriction of hazardous substances, EU market)
- REACH (Chemical compliance, EU market)

3. MATERIAL REQUIREMENTS

Base Material: Aluminum Alloy 6061-T6 per ASTM B209

Surface Finish: Hard anodize 25 micron min, MIL-A-8625 Type III

Corrosion: Salt spray 500 hours per ASTM B117

Alternate alloys: ADC12, A380, LM25, AlSi9Cu3 " prior approval needed

4. DIMENSIONAL REQUIREMENTS

Overall Envelope: L 180mm x W 120mm x H 95mm
Wall Thickness: 3.5mm +/- 0.2mm
Bore Diameter: 52.000mm +0.025/-0.000mm (H7 fit)
Mounting Hole PCD: 100mm +/- 0.05mm
Surface Roughness: Ra 1.6 max on sealing surfaces
Flatness: 0.05mm on mating faces

5. MANDATORY CERTIFICATION REQUIREMENTS

REQUIRED: ISO 9001:2015 (quality management system)
PREFERRED: AS9100D (for aerospace-grade variants only)
REQUIRED: RoHS Compliant (all variants shipped to EU customers)
Expired certifications: NOT accepted. Suppliers must provide current cert.

6. SUPPLIER QUALIFICATION

First Article Inspection (FAI): Required for all new or re-qualified suppliers
Incoming Inspection AQL: 1.0 (critical dims), 2.5 (major features)
CMM Dimensional Report: Required with each production lot
Material Test Certificate: Required per lot (heat number traceability)
Annual Supplier Audit: Required for Preferred Supplier status

7. DISQUALIFICATION CONDITIONS

- ISO 9001:2015 not held OR expired
- Defect rate > 10% on any previous contract
- Failed FAI without corrective action within 30 days
- Single-source dependencies without approved contingency plan
- Country risk score below 60/100 (procurement policy PR-2026-04)

APPENDIX A – SUPPLIER QUALIFICATION AND APPROVAL PROCESS

A1. NEW SUPPLIER QUALIFICATION

- Step 1: Supplier Self-Assessment Questionnaire (SAQ) – returned within 10 days
- Step 2: Quality Management System documentation review
- Step 3: On-site audit (for orders > USD 50,000/year or safety-critical parts)
- Step 4: First Article Inspection (FAI) per AS9102 or equivalent
- Step 5: Production Part Approval Process (PPAP) Level 3 minimum
- Step 6: Approved Supplier List (ASL) entry – valid 3 years with annual review

A2. FIRST ARTICLE INSPECTION REQUIREMENTS

- Balloon drawing with all dimensions inspected and recorded
- Material test certificate with heat number traceability
- Surface finish measurement report (Ra on all critical surfaces)
- Functional test report where applicable
- Dimensional report: CMM or conventional measurement, all GD&T callouts

A3. ONGOING SUPPLIER PERFORMANCE METRICS

- On-Time Delivery (OTD): Target $\geq 95\%$ | Minimum acceptable: 90%
- Incoming Quality Rate (IQR): Target $\leq 2\%$ defect | Disqualify: $> 10\%$
- Corrective Action Response: ≤ 10 business days for 8D response
- Customer Satisfaction Score: Tracked quarterly via Supplier Scorecard
- Annual Business Review: Mandatory for all suppliers > USD 100K/year

A4. DISQUALIFICATION TRIGGERS

- Lapse of mandatory certification (ISO 9001:2015 or equivalent)
- Three consecutive OTD failures below 90%
- Single-lot defect rate exceeding 10% incoming AQL
- Failure to respond to Corrective Action Request within 30 days
- Fraudulent documentation, sub-tier substitution without approval
- Country risk rating below threshold per procurement policy PR-2026-04

A5. APPROVED TESTING LABORATORIES

- For RoHS/REACH: Bureau Veritas, Intertek, SGS, TUV SUD, Eurofins
- For dimensional: Any ISO/IEC 17025 accredited CMM facility
- For material: Mill certificate + independent 3rd party for critical alloys

A6. PACKAGING AND LABELING REQUIREMENTS

- Individual part protection: Bubble wrap or foam tray, no bare metal contact
- Inner carton: Max 20 units, moisture-desiccant included
- Outer carton: Double-wall corrugated, gross weight max 20 kg
- Label (minimum): Part number, revision, quantity, date code, supplier ID, PO#
- Country of origin marking: Per importing country requirements
- Special handling: Fragile, this side up, humidity-sensitive as applicable